$$s = \lim_{n\to\infty} 2^n s_n \tag{1a}$$
$$s_n = \sqrt{r^2 + r^2 - 2ra_n} \tag{1b}$$
$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \tag{1c}$$
$$a_0 = x \quad \text{for } -r \le x \le r \tag{1d}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{r^2 + r^2 - 2ra_n} \tag{2a}$$
$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \tag{2b}$$
$$a_0 = x \quad \text{for } -r \le x \le r \tag{2c}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{3a}$$
$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \tag{3b}$$
$$a_0 = x \quad \text{for } -r \le x \le r \tag{3c}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{4a}$$
$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \tag{4b}$$
$$a_0 = x \quad \text{for } -r \le x \le r \tag{4c}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{5a}$$
$$a_n = r\frac{\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \tag{5b}$$
$$a_0 = x \quad \text{for } -r \le x \le r \tag{5c}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{6a}$$
$$a_n = r\sqrt{\frac{r^2 - a_{n-1}^2}{2r^2 - 2ra_{n-1}}} \tag{6b}$$
$$a_0 = x \quad \text{for } -r \le x \le r \tag{6c}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{7a}$$
$$a_n = r\sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r^2 - 2ra_{n-1}}} \tag{7b}$$
$$a_0 = x \quad \text{for } -r \le x \le r \tag{7c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{8a}$$

$$a_n = r\sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r(r - a_{n-1})}} \tag{8b}$$

$$a_0 = x \quad \text{for } -r \le x \le r \tag{8c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{9a}$$

$$a_n = r\sqrt{\frac{r + a_{n-1}}{2r}} \tag{9b}$$

$$a_0 = x \quad \text{for } -r \le x \le r \tag{9c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{10a}$$

$$a_n = r\sqrt{\frac{2r(r + a_{n-1})}{(2r)^2}} \tag{10b}$$

$$a_0 = x \quad \text{for } -r \le x \le r \tag{10c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{11a}$$

$$a_n = r\sqrt{\frac{2r^2 + 2ra_{n-1}}{(2r)^2}} \tag{11b}$$

$$a_0 = x \quad \text{for } -r \le x \le r \tag{11c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{12a}$$

$$a_n = r\sqrt{\frac{\left(\sqrt{2r^2 + 2ra_{n-1}}\right)^2}{(2r)^2}} \tag{12b}$$

$$a_0 = x \quad \text{for } -r \le x \le r \tag{12c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{13a}$$

$$a_n = r\sqrt{\left(\frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}\right)^2} \tag{13b}$$

$$a_0 = x \quad \text{for } -r \le x \le r \tag{13c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{14a}$$

$$a_n = r\frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r} \tag{14b}$$

$$a_0 = x \quad \text{for } -r \le x \le r \tag{14c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{15a}$$
$$a_n = \frac{r\sqrt{2r^2 + 2ra_{n-1}}}{2r} \tag{15b}$$
$$a_0 = x \quad \text{for } -r \leq x \leq r \tag{15c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{16a}$$
$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \tag{16b}$$
$$a_0 = x \quad \text{for } -r \leq x \leq r \tag{16c}$$

$$A_{ORS_{n-1}} = \frac{\overline{RS_{n-1}OX_n}}{2} = \frac{\overline{OS_{n-1}RX_{n-1}}}{2} \tag{17}$$
$$\frac{\overline{RS_{n-1}OX_n}}{2} = \frac{\overline{OS_{n-1}RX_{n-1}}}{2} \tag{18}$$
$$\overline{RS_{n-1}OX_n} = \overline{OS_{n-1}RX_{n-1}} \tag{19}$$
$$\frac{\overline{RS_{n-1}OX_n}}{\overline{RS_{n-1}}} = \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}} \tag{20}$$
$$\overline{OX_n} = \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}} \tag{21}$$
$$\overline{OX_n} = \frac{\overline{OS_{n-1}}\sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\overline{RS_{n-1}}} \tag{22}$$
$$\overline{OX_n} = \frac{\overline{OS_{n-1}}\sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\sqrt{\overline{OR}^2 + \overline{OS_{n-1}}^2 - 2\overline{OS_{n-1}OX_{n-1}}}} \tag{23}$$
$$\overline{RS_n}^2 = \overline{RX_n}^2 + \overline{X_nS_n}^2 \tag{24}$$
$$\overline{RS_n} = \sqrt{\overline{RX_n}^2 + \overline{X_nS_n}^2} \tag{25}$$
$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{X_nS_n}^2} \tag{26}$$
$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \left(\overline{OS_n} - \overline{OX_n}\right)^2} \tag{27}$$
$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{OS_n}^2 - 2\overline{OS_nOX_n} + \overline{OX_n}^2} \tag{28}$$
$$\overline{RS_n} = \sqrt{\overline{OR}^2 + \overline{OS_n}^2 - 2\overline{OS_nOX_n}} \tag{29}$$
$$\overline{OX_n}^2 + \overline{RX_n}^2 = \overline{OR}^2 \tag{30}$$
$$\overline{OX_n}^2 + \overline{RX_n}^2 - \overline{OX_n}^2 = \overline{OR}^2 - \overline{OX_n}^2 \tag{31}$$
$$\overline{RX_n}^2 = \overline{OR}^2 - \overline{OX_n}^2 \tag{32}$$
$$\overline{RX_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2} \tag{33}$$

$$A = \frac{rs}{2} \tag{34}$$

$$A_0 = \lim_{n\to\infty} 2^{n-1} A_{ORS_nS_{n-1}} \tag{35}$$

$$A_0 = \lim_{n\to\infty} 2^{n-1} \frac{\overline{RS_{n-1}OS_n}}{2} \tag{36}$$

$$A_0 = \frac{\lim\limits_{n\to\infty} 2^{n-1}\overline{RS_{n-1}OS_n}}{2} \tag{37}$$

$$A_0 = \frac{\lim\limits_{n\to\infty} \left(2^{n-1}\overline{RS_{n-1}}\right)\overline{OS_n}}{2} \tag{38}$$

$$A_0 = \frac{\lim\limits_{n\to\infty} \overline{OS_n}\left(2^{n-1}\overline{RS_{n-1}}\right)}{2} \tag{39}$$

$$A_0 = \frac{\left(\lim\limits_{n\to\infty} \overline{OS_n}\right)\left(\lim\limits_{n\to\infty} 2^{n-1}\overline{RS_{n-1}}\right)}{2} \tag{40}$$

$$A_0 = \frac{\overline{OR}\lim\limits_{n\to\infty} 2^{n-1}\overline{RS_{n-1}}}{2} \tag{41}$$

$$A_0 = \frac{\overline{OR}\,\mathrm{arc}\,RS_0}{2} \tag{42}$$

$$\begin{aligned} \arccos\colon \{(r,x) \in \mathbb{R}^2 \mid r > 0, -r \le x \le r\} &\to \mathbb{R} \\ (r,x) &\mapsto s \end{aligned} \tag{43}$$

$$\arccos(r,x) = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{44a}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \tag{44b}$$

$$a_0 = x \tag{44c}$$

$$\begin{aligned} \cos\colon \{(r,s) \in \mathbb{R}^2 \mid r > 0\} &\to \mathbb{R} \\ (r,s) &\mapsto x \end{aligned} \tag{45}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{46a}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \tag{46b}$$

$$a_0 = \cos(r,s) \tag{46c}$$

$$s = \lim_{m\to\infty} 2^m \sqrt{2r^2 - 2ra_{m,m}} \tag{47a}$$

$$a_{m,n} = \frac{\sqrt{2r^2 + 2ra_{m,n-1}}}{2} \quad \text{for } n \le m \tag{47b}$$

$$a_{m,0} = \cos(r,s) \tag{47c}$$

$$\cos(r, s) = \lim_{m\to\infty} a_{m,m} \tag{48a}$$

$$a_{m,n-1} = \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \tag{48b}$$

$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{48c}$$

$$\cos(r, s) = \lim_{m\to\infty} a_{m,m} \tag{49a}$$

$$2a_{m,n-1} = 2\frac{\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \tag{49b}$$

$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{49c}$$

$$\cos(r, s) = \lim_{m\to\infty} a_{m,m} \tag{50a}$$

$$2a_{m,n-1} = \frac{2\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \tag{50b}$$

$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{50c}$$

$$\cos(r, s) = \lim_{m\to\infty} a_{m,m} \tag{51a}$$

$$2a_{m,n-1} = \sqrt{2r^2 + 2ra_{m,n}} \quad \text{for } n \leq m \tag{51b}$$

$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{51c}$$

$$\cos(r, s) = \lim_{m\to\infty} a_{m,m} \tag{52a}$$

$$(2a_{m,n-1})^2 = \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2 \quad \text{for } n \leq m \tag{52b}$$

$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{52c}$$

$$\cos(r, s) = \lim_{m\to\infty} a_{m,m} \tag{53a}$$

$$2^2a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2 \quad \text{for } n \leq m \tag{53b}$$

$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{53c}$$

$$\cos(r, s) = \lim_{m\to\infty} a_{m,m} \tag{54a}$$

$$4a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2 \quad \text{for } n \leq m \tag{54b}$$

$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{54c}$$

$$\cos(r, s) = \lim_{m\to\infty} a_{m,m} \tag{55a}$$

$$4a_{m,n-1}^2 = 2r^2 + 2ra_{m,n} \quad \text{for } n \leq m \tag{55b}$$

$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{55c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{56a}$$
$$4a_{m,n-1}^2 - 2r^2 = 2r^2 + 2ra_{m,n} - 2r^2 \quad \text{for } n \le m \tag{56b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{56c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{57a}$$
$$4a_{m,n-1}^2 - 2r^2 = 2ra_{m,n} \quad \text{for } n \le m \tag{57b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{57c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{58a}$$
$$\frac{4a_{m,n-1}^2 - 2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \le m \tag{58b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{58c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{59a}$$
$$\frac{4a_{m,n-1}^2}{2r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \le m \tag{59b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{59c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{60a}$$
$$\frac{2a_{m,n-1}^2}{r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \le m \tag{60b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{60c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{61a}$$
$$\frac{2a_{m,n-1}^2}{r} - r = \frac{2ra_{m,n}}{2r} \quad \text{for } n \le m \tag{61b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{61c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{62a}$$
$$\frac{2a_{m,n-1}^2}{r} - r = a_{m,n} \quad \text{for } n \le m \tag{62b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{62c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{63a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \le m \tag{63b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s \tag{63c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{64a}$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{64b}$$

$$\frac{2^m\sqrt{2r^2 - 2ra_{m,0}}}{2^m} = \frac{s}{2^m} \tag{64c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{65a}$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{65b}$$

$$\sqrt{2r^2 - 2ra_{m,0}} = \frac{s}{2^m} \tag{65c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{66a}$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{66b}$$

$$\left(\sqrt{2r^2 - 2ra_{m,0}}\right)^2 = \left(\frac{s}{2^m}\right)^2 \tag{66c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{67a}$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{67b}$$

$$2r^2 - 2ra_{m,0} = \left(\frac{s}{2^m}\right)^2 \tag{67c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{68a}$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{68b}$$

$$2r^2 - 2ra_{m,0} = \frac{s^2}{(2^m)^2} \tag{68c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{69a}$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{69b}$$

$$2r^2 - 2ra_{m,0} = \frac{s^2}{2^{2m}} \tag{69c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{70a}$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{70b}$$

$$2r^2 - 2ra_{m,0} - 2r^2 = \frac{s^2}{2^{2m}} - 2r^2 \tag{70c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{71a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{71b}$$
$$-2ra_{m,0} = \frac{s^2}{2^{2m}} - 2r^2 \tag{71c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{72a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{72b}$$
$$\frac{-2ra_{m,0}}{-2r} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r} \tag{72c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{73a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{73b}$$
$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r} \tag{73c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{74a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{74b}$$
$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right)}{-2r} - \frac{2r^2}{-2r} \tag{74c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{75a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{75b}$$
$$a_{m,0} = \frac{s^2}{2^{2m}(-2r)} - \frac{2r^2}{-2r} \tag{75c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{76a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{76b}$$
$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} - \frac{2r^2}{-2r} \tag{76c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{77a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \tag{77b}$$
$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + \frac{-2r^2}{-2r} \tag{77c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{78a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \le m \tag{78b}$$
$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + r \tag{78c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{79a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \le m \tag{79b}$$
$$a_{m,0} = r + \frac{s^2}{-2^{2m+1}r} \tag{79c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{80a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \le m \tag{80b}$$
$$a_{m,0} = r - \left(-\frac{s^2}{-2^{2m+1}r}\right) \tag{80c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{81a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \le m \tag{81b}$$
$$a_{m,0} = r - \left(\frac{-s^2}{-2^{2m+1}r}\right) \tag{81c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{82a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \le m \tag{82b}$$
$$a_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{82c}$$

$$\cos(r,s) = \lim_{m\to\infty} ra_{m,m} \tag{83a}$$
$$ra_{m,n} = \frac{2(ra_{m,n-1})^2}{r} - r \quad \text{for } n \le m \tag{83b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{83c}$$

$$\cos(r,s) = r\lim_{m\to\infty} a_{m,m} \tag{84a}$$
$$ra_{m,n} = \frac{2(ra_{m,n-1})^2}{r} - r \quad \text{for } n \le m \tag{84b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{84c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{85a}$$
$$ra_{m,n} = \frac{2r^2 a_{m,n-1}^2}{r} - r \quad \text{for } n \le m \tag{85b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{85c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{86a}$$
$$ra_{m,n} = 2ra_{m,n-1}^2 - r \quad \text{for } n \le m \tag{86b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{86c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{87a}$$
$$\frac{ra_{m,n}}{r} = \frac{2ra_{m,n-1}^2 - r}{r} \quad \text{for } n \le m \tag{87b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{87c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{88a}$$
$$\frac{ra_{m,n}}{r} = \frac{2ra_{m,n-1}^2}{r} - \frac{r}{r} \quad \text{for } n \le m \tag{88b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{88c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{89a}$$
$$a_{m,n} = \frac{2ra_{m,n-1}^2}{r} - \frac{r}{r} \quad \text{for } n \le m \tag{89b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{89c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{90a}$$
$$a_{m,n} = 2a_{m,n-1}^2 - \frac{r}{r} \quad \text{for } n \le m \tag{90b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{90c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{91a}$$
$$a_{m,n} = 2a_{m,n-1}^2 - 1 \quad \text{for } n \le m \tag{91b}$$
$$ra_{m,0} = r - \frac{s^2}{2^{2m+1}r} \tag{91c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{92a}$$
$$a_{m,n} = 2a_{m,n-1}^2 - 1 \quad \text{for } n \leq m \tag{92b}$$
$$\frac{ra_{m,0}}{r} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r} \tag{92c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{93a}$$
$$a_{m,n} = 2a_{m,n-1}^2 - 1 \quad \text{for } n \leq m \tag{93b}$$
$$\frac{ra_{m,0}}{r} = \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r} \tag{93c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{94a}$$
$$a_{m,n} = 2a_{m,n-1}^2 - 1 \quad \text{for } n \leq m \tag{94b}$$
$$a_{m,0} = \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r} \tag{94c}$$

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$$a_{m,0} = 1 - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r} \tag{95c}$$

$$\cos(r,s) = r \lim_{m\to\infty} a_{m,m} \tag{96a}$$
$$a_{m,n} = 2a_{m,n-1}^2 - 1 \quad \text{for } n \leq m \tag{96b}$$
$$a_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2} \tag{96c}$$

$$T_{2^n}(x) = T_{2(2^{n-1})}(x) \tag{97}$$

$$T_{2^n}(x) = 2(T_{2^{n-1}}(x))^2 - 1 \tag{98a}$$
$$T_1(x) = x \tag{98b}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}(x-1)^k \tag{99}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{2^k (x-1)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!} \tag{100}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2(x-1))^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!} \tag{101}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \tag{102}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} (2^{2n} - j^2) \tag{103}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} \frac{2^{2n}(2^{2n} - j^2)}{2^{2n}} \tag{104}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} 2^{2n} \frac{2^{2n} - j^2}{2^{2n}} \tag{105}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} 2^{2n} \prod_{j=0}^{k-1} \frac{2^{2n} - j^2}{2^{2n}} \tag{106}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2(x-1))^k}{(2k)!} (2^{2n})^k \prod_{j=0}^{k-1} \frac{2^{2n} - j^2}{2^{2n}} \tag{107}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2(x-1))^k (2^{2n})^k}{(2k)!} \prod_{j=0}^{k-1} \frac{2^{2n} - j^2}{2^{2n}} \tag{108}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2^{2n+1}(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} \frac{2^{2n} - j^2}{2^{2n}} \tag{109}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2^{2n+1}(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} \left(\frac{2^{2n}}{2^{2n}} - \frac{j^2}{2^{2n}} \right) \tag{110}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2^{2n+1}(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{2^{2n}} \right) \tag{111}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2^{2n+1}(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{(2^2)^n} \right) \tag{112}$$

$$T_{2^n}(x) = \sum_{k=0}^{2^n} \frac{(2^{2n+1}(x-1))^k}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^n} \right) \tag{113}$$

$$\cos(r, s) = r \lim_{n\to\infty} T_{2^n} \left(1 - \frac{s^2}{2^{2n+1} r^2} \right) \tag{114}$$

$$\cos(r, s) = r \lim_{n\to\infty} \sum_{k=0}^{2^n} \frac{\left(2^{2n+1} \left(1 - \frac{s^2}{2^{2n+1} r^2} - 1\right)\right)^k}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^n} \right) \tag{115}$$

$$\cos(r, s) = r \lim_{n\to\infty} \sum_{k=0}^{2^n} \frac{\left(2^{2n+1} \left(- \frac{s^2}{2^{2n+1} r^2}\right)\right)^k}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^n} \right) \tag{116}$$

$$\cos(r,s) = r\lim_{n\to\infty}\sum_{k=0}^{2^n}\frac{\left(-2^{2n+1}\frac{s^2}{2^{2n+1}r^2}\right)^k}{(2k)!}\prod_{j=0}^{k-1}\left(1-\frac{j^2}{4^n}\right) \tag{117}$$

$$\cos(r,s) = r\lim_{n\to\infty}\sum_{k=0}^{2^n}\frac{\left(-\frac{2^{2n+1}s^2}{2^{2n+1}r^2}\right)^k}{(2k)!}\prod_{j=0}^{k-1}\left(1-\frac{j^2}{4^n}\right) \tag{118}$$

$$\cos(r,s) = r\lim_{n\to\infty}\sum_{k=0}^{2^n}\frac{\left(-\frac{s^2}{r^2}\right)^k}{(2k)!}\prod_{j=0}^{k-1}\left(1-\frac{j^2}{4^n}\right) \tag{119}$$

$$\cos(r,s) = r\lim_{n\to\infty}\sum_{k=0}^{2^n}\frac{(-1)^k\left(\frac{s^2}{r^2}\right)^k}{(2k)!}\prod_{j=0}^{k-1}\left(1-\frac{j^2}{4^n}\right) \tag{120}$$

$$\cos(r,s) = r\lim_{n\to\infty}\sum_{k=0}^{2^n}\frac{(-1)^k\left(\left(\frac{s}{r}\right)^2\right)^k}{(2k)!}\prod_{j=0}^{k-1}\left(1-\frac{j^2}{4^n}\right) \tag{121}$$

$$\cos(r,s) = r\lim_{n\to\infty}\sum_{k=0}^{2^n}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}\left(1-\frac{j^2}{4^n}\right) \tag{122}$$

$$\cos(r,s) = r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{123}$$

$$\begin{aligned}\arcsin\colon \{(r,y)\in\mathbb{R}^2 \mid r>0, -r\le y\le r\} &\to \mathbb{R}\\ (r,y) &\mapsto s\end{aligned} \tag{124}$$

$$\arcsin(r,y) = \frac{\arcsin(r,y)+\arcsin(r,y)}{2} \tag{125}$$

$$\arcsin(r,y) = \frac{\arcsin(r,y)+\arcsin(r,y)+\arccos(r,y)-\arccos(r,y)}{2} \tag{126}$$

$$\arcsin(r,y) = \frac{\arcsin(r,y)+\arcsin(r,-y)+\arccos(r,-y)-\arccos(r,y)}{2} \tag{127}$$

$$\arcsin(r,y) = \frac{\arcsin(r,y)-\arcsin(r,y)+\arccos(r,-y)-\arccos(r,y)}{2} \tag{128}$$

$$\arcsin(r,y) = \frac{\arccos(r,-y)-\arccos(r,y)}{2} \tag{129}$$

$$\begin{aligned}\sin\colon \{(r,s)\in\mathbb{R}^2 \mid r>0\} &\to \mathbb{R}\\ (r,s) &\mapsto y\end{aligned} \tag{130}$$

$$\sin(r,s) = r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{131}$$

$$\cos(r, \alpha \pm \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha \pm \beta}{r}\right)^{2k}}{(2k)!} \tag{132}$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \to \infty} \sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha \pm \beta}{r}\right)^{2k}}{(2k)!} \tag{133}$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \to \infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k} \binom{2k}{j} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{\pm \beta}{r}\right)^{j}}{(2k)!} \tag{134}$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \to \infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k} \frac{(2k)!}{j!(2k-j)!} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{\pm \beta}{r}\right)^{j}}{(2k)!} \tag{135}$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \to \infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k} (2k)! \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{\pm \beta}{r}\right)^{j}}{j!(2k-j)!}}{(2k)!} \tag{136}$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \to \infty} \sum_{k=0}^{n} \frac{(-1)^k (2k)! \sum_{j=0}^{2k} \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{\pm \beta}{r}\right)^{j}}{j!(2k-j)!}}{(2k)!} \tag{137}$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \to \infty} \sum_{k=0}^{n} (-1)^k \sum_{j=0}^{2k} \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{\pm \beta}{r}\right)^{j}}{j!(2k-j)!} \tag{138}$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \to \infty} \left(\sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j}}{(2j)!} \mp \sum_{k=0}^{n-1} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \tag{139}$$

$$\cos(r, \alpha \pm \beta) = r \left(\lim_{n \to \infty} \left(\sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j}}{(2j)!} \right) \mp \lim_{n \to \infty} \left(\sum_{k=0}^{n-1} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \right) \tag{140}$$

$$\cos(r,\alpha\pm\beta)=r\left(\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\mp\lim_{n\to\infty}\left(\sum_{k=0}^{n-1}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{j=0}^{n-1-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!}\right)\right) \tag{141}$$

$$\cos(r,\alpha\pm\beta)=r\left(\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\mp\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right) \tag{142}$$

$$\cos(r,\alpha\pm\beta)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\mp r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \tag{143}$$

$$\cos(r,\alpha\pm\beta)=\frac{r\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\mp r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{144}$$

$$\cos(r,\alpha\pm\beta)=\frac{r^2\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\mp r^2\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{145}$$

$$\cos(r,\alpha\pm\beta)=\frac{\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)\mp r^2\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{146}$$

$$\cos(r,\alpha\pm\beta)=\frac{\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)\mp\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{147}$$

$$\cos(r,\alpha\pm\beta)=\frac{\cos(r,\alpha)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)\mp\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{148}$$

$$\cos(r,\alpha\pm\beta)=\frac{\cos(r,\alpha)\cos(r,\beta)\mp\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{149}$$

$$\cos(r,\alpha-\beta) = a_{0,0} \tag{150a}$$

$$a_{0,0} = r - \frac{\left(\sqrt{\left(\cos(r,\alpha)-\cos(r,\beta)\right)^2 + \left(\sin(r,\alpha)-\sin(r,\beta)\right)^2}\right)^2}{2r} \tag{150b}$$

$$\cos(r,\alpha-\beta) = r - \frac{\left(\sqrt{\left(\cos(r,\alpha)-\cos(r,\beta)\right)^2 + \left(\sin(r,\alpha)-\sin(r,\beta)\right)^2}\right)^2}{2r} \tag{151}$$

$$\cos(r,\alpha-\beta) = r - \frac{\left(\cos(r,\alpha)-\cos(r,\beta)\right)^2 + \left(\sin(r,\alpha)-\sin(r,\beta)\right)^2}{2r} \tag{152}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2}{2r} - \frac{\left(\cos(r,\alpha)-\cos(r,\beta)\right)^2 + \left(\sin(r,\alpha)-\sin(r,\beta)\right)^2}{2r} \tag{153}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2 - \left(\left(\cos(r,\alpha)-\cos(r,\beta)\right)^2 + \left(\sin(r,\alpha)-\sin(r,\beta)\right)^2\right)}{2r} \tag{154}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2 - \left(\cos(r,\alpha)-\cos(r,\beta)\right)^2 - \left(\sin(r,\alpha)-\sin(r,\beta)\right)^2}{2r} \tag{155}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2 - (\cos^2(r,\alpha) - 2\cos(r,\alpha)\cos(r,\beta) + \cos^2(r,\beta)) - \left(\sin(r,\alpha)-\sin(r,\beta)\right)^2}{2r} \tag{156}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2 - (\cos^2(r,\alpha) - 2\cos(r,\alpha)\cos(r,\beta) + \cos^2(r,\beta)) - \left(\sin^2(r,\alpha) - 2\sin(r,\alpha)\sin(r,\beta) + \sin^2(r,\beta)\right)}{2r} \tag{157}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2 - \cos^2(r,\alpha) + 2\cos(r,\alpha)\cos(r,\beta) - \cos^2(r,\beta) - \left(\sin^2(r,\alpha) - 2\sin(r,\alpha)\sin(r,\beta) + \sin^2(r,\beta)\right)}{2r} \tag{158}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2 - \cos^2(r,\alpha) + 2\cos(r,\alpha)\cos(r,\beta) - \cos^2(r,\beta) - \sin^2(r,\alpha) + 2\sin(r,\alpha)\sin(r,\beta) - \sin^2(r,\beta}{2r} \tag{159}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2 - \left(\cos^2(r,\alpha) + \sin^2(r,\alpha)\right) + 2\cos(r,\alpha)\cos(r,\beta) - \cos^2(r,\beta) + 2\sin(r,\alpha)\sin(r,\beta) - \sin^2(r,\beta}{2r} \tag{160}$$

$$\cos(r,\alpha-\beta) = \frac{2r^2 - \left(\cos^2(r,\alpha) + \sin^2(r,\alpha)\right) - \left(\cos^2(r,\beta) + \sin^2(r,\beta)\right) + 2\cos(r,\alpha)\cos(r,\beta) + 2\sin(r,\alpha)\sin(r,\beta}{2r} \tag{161}$$

$$\cos(r,\alpha-\beta)=\frac{2r^2-\left(\cos^2(r,\alpha)+\sin^2(r,\alpha)\right)-\left(\cos^2(r,\beta)+\sin^2(r,\beta)\right)+2\left(\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)\right)}{2r} \tag{162}$$

$$\cos(r,\alpha-\beta)=\frac{2r^2-r^2-\left(\cos^2(r,\beta)+\sin^2(r,\beta)\right)+2\left(\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)\right)}{2r} \tag{163}$$

$$\cos(r,\alpha-\beta)=\frac{r^2-\left(\cos^2(r,\beta)+\sin^2(r,\beta)\right)+2\left(\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)\right)}{2r} \tag{164}$$

$$\cos(r,\alpha-\beta)=\frac{r^2-r^2+2\left(\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)\right)}{2r} \tag{165}$$

$$\cos(r,\alpha-\beta)=\frac{2\left(\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)\right)}{2r} \tag{166}$$

$$\cos(r,\alpha-\beta)=\frac{\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)}{r} \tag{167}$$

$$\sin(r,\alpha)\sin(r,\beta)=\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right) \tag{168}$$

$$\sin(r,\alpha)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \tag{169}$$

$$\sin(r,\beta)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \tag{170}$$

$$\cos(r,\alpha\pm\beta)=\frac{\cos(r,\alpha)\cos(r,\beta)\mp\sin(r,\alpha)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{171}$$

$$\cos(r,\alpha\pm\beta)=\frac{\cos(r,\alpha)\cos(r,\beta)\mp\sin(r,\alpha)\sin(r,\beta)}{r} \tag{172}$$

$$\sin(r,\alpha\pm\beta)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha\pm\beta}{r}\right)^{2k+1}}{(2k+1)!} \tag{173}$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha\pm\beta}{r}\right)^{2k+1}}{(2k+1)!} \tag{174}$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k+1} \binom{2k+1}{j} \left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm\beta}{r}\right)^{j}}{(2k+1)!} \tag{175}$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k+1} \frac{(2k+1)!}{j!(2k+1-j)!} \left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm\beta}{r}\right)^{j}}{(2k+1)!} \tag{176}$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k+1} (2k+1)! \frac{\left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm\beta}{r}\right)^{j}}{j!(2k+1-j)!}}{(2k+1)!} \tag{177}$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k (2k+1)! \sum_{j=0}^{2k+1} \frac{\left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm\beta}{r}\right)^{j}}{j!(2k+1-j)!}}{(2k+1)!} \tag{178}$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} (-1)^k \sum_{j=0}^{2k+1} \frac{\left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm\beta}{r}\right)^{j}}{j!(2k+1-j)!} \tag{179}$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n\to\infty} \left(\sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j}}{(2j)!} \pm \sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \tag{180}$$

$$\sin(r, \alpha \pm \beta) = r \left(\lim_{n\to\infty} \left(\sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j}}{(2j)!} \right) \pm \lim_{n\to\infty} \left(\sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \right) \tag{181}$$

$$\sin(r, \alpha \pm \beta) = r \left(\sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \pm \lim_{n\to\infty} \left(\sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \right) \tag{182}$$

$$\sin(r,\alpha\pm\beta)=r\left(\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\pm\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right) \tag{183}$$

$$\sin(r,\alpha\pm\beta)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\pm r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \tag{184}$$

$$\sin(r,\alpha\pm\beta)=\frac{r\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\pm r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{185}$$

$$\sin(r,\alpha\pm\beta)=\frac{r^2\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\pm r^2\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{186}$$

$$\sin(r,\alpha\pm\beta)=\frac{\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)\pm r^2\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{187}$$

$$\sin(r,\alpha\pm\beta)=\frac{\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)\pm\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{188}$$

$$\sin(r,\alpha\pm\beta)=\frac{\sin(r,\alpha)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)\pm\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{189}$$

$$\sin(r,\alpha\pm\beta)=\frac{\sin(r,\alpha)\cos(r,\beta)\pm\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{190}$$

$$\sin(r,\alpha\pm\beta)=\frac{\sin(r,\alpha)\cos(r,\beta)\pm\cos(r,\alpha)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{191}$$

$$\sin(r,\alpha\pm\beta)=\frac{\sin(r,\alpha)\cos(r,\beta)\pm\cos(r,\alpha)\sin(r,\beta)}{r} \tag{192}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \right)}{\partial s} \tag{193}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k}}{r^{2k}}}{(2k)!} \right)}{\partial s} \tag{194}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k}(2k)!} \right)}{\partial s} \tag{195}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k}}{r^{2k}(2k)!} \right)}{\partial s} \tag{196}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k-1}(2k)!} \right)}{\partial s} \tag{197}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k}}{\partial s}}{r^{2k-1}(2k)!} \tag{198}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) s^{2k-1}}{r^{2k-1}(2k)!} \tag{199}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \frac{s^{2k-1}}{r^{2k-1}}}{(2k)!} \tag{200}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)!} \tag{201}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)(2k-1)!} \tag{202}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{203}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{-(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{204}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{205}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2k-1)!} \tag{206}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2(k-1)+1)!} \tag{207}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{208}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{-r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{209}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{-\sin(r,s)}{r} \tag{210}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \right)}{\partial s} \tag{211}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k+1}}{r^{2k+1}}}{(2k+1)!} \right)}{\partial s} \tag{212}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k+1}(2k+1)!} \right)}{\partial s} \tag{213}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k+1}}{r^{2k+1}(2k+1)!} \right)}{\partial s} \tag{214}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k}(2k+1)!} \right)}{\partial s} \tag{215}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k+1}}{\partial s}}{r^{2k}(2k+1)!} \tag{216}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) s^{2k}}{r^{2k}(2k+1)!} \tag{217}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \frac{s^{2k}}{r^{2k}}}{(2k+1)!} \tag{218}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)!} \tag{219}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)(2k)!} \tag{220}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{221}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}}{r} \tag{222}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\cos(r,s)}{r} \tag{223}$$

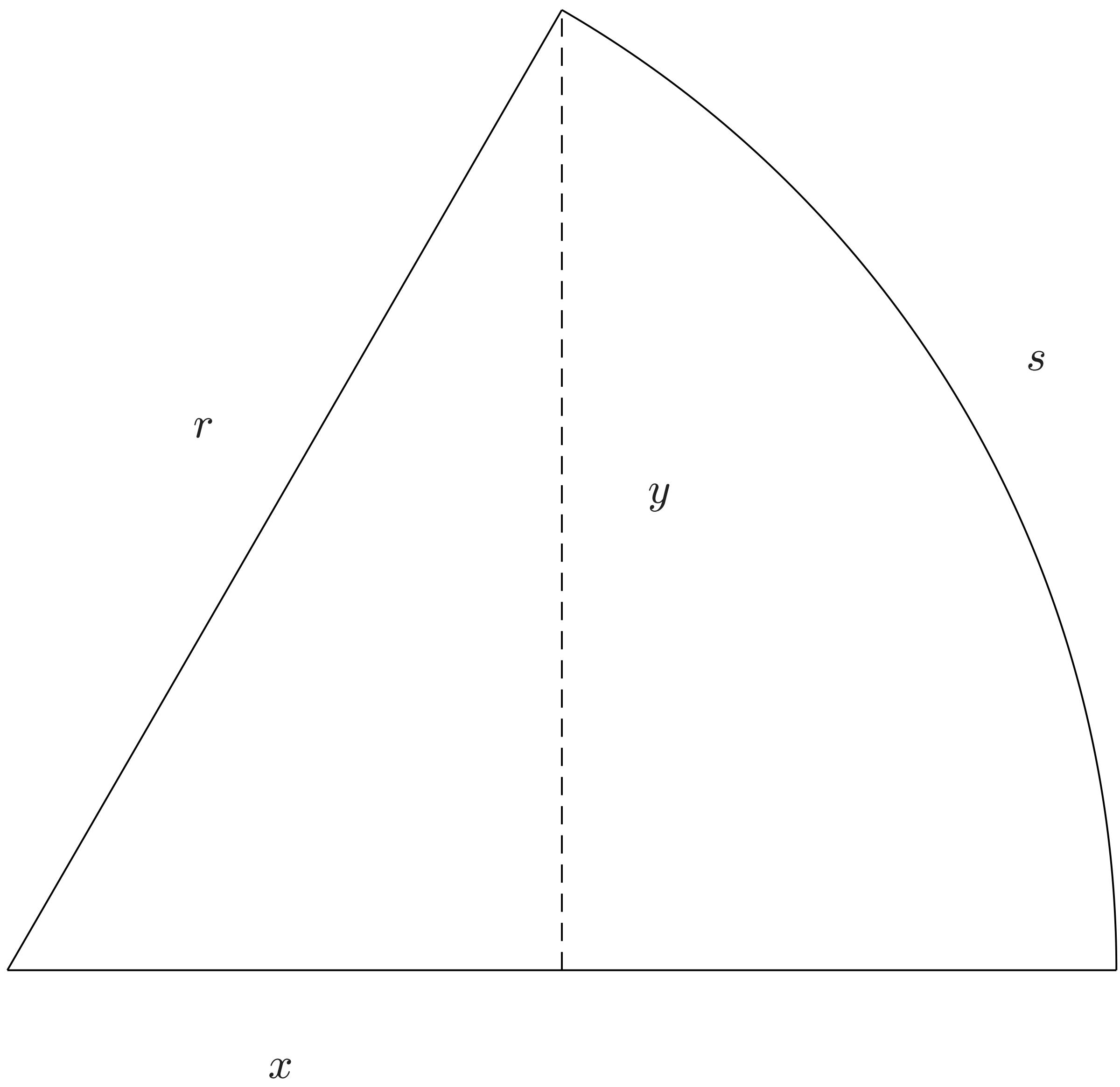
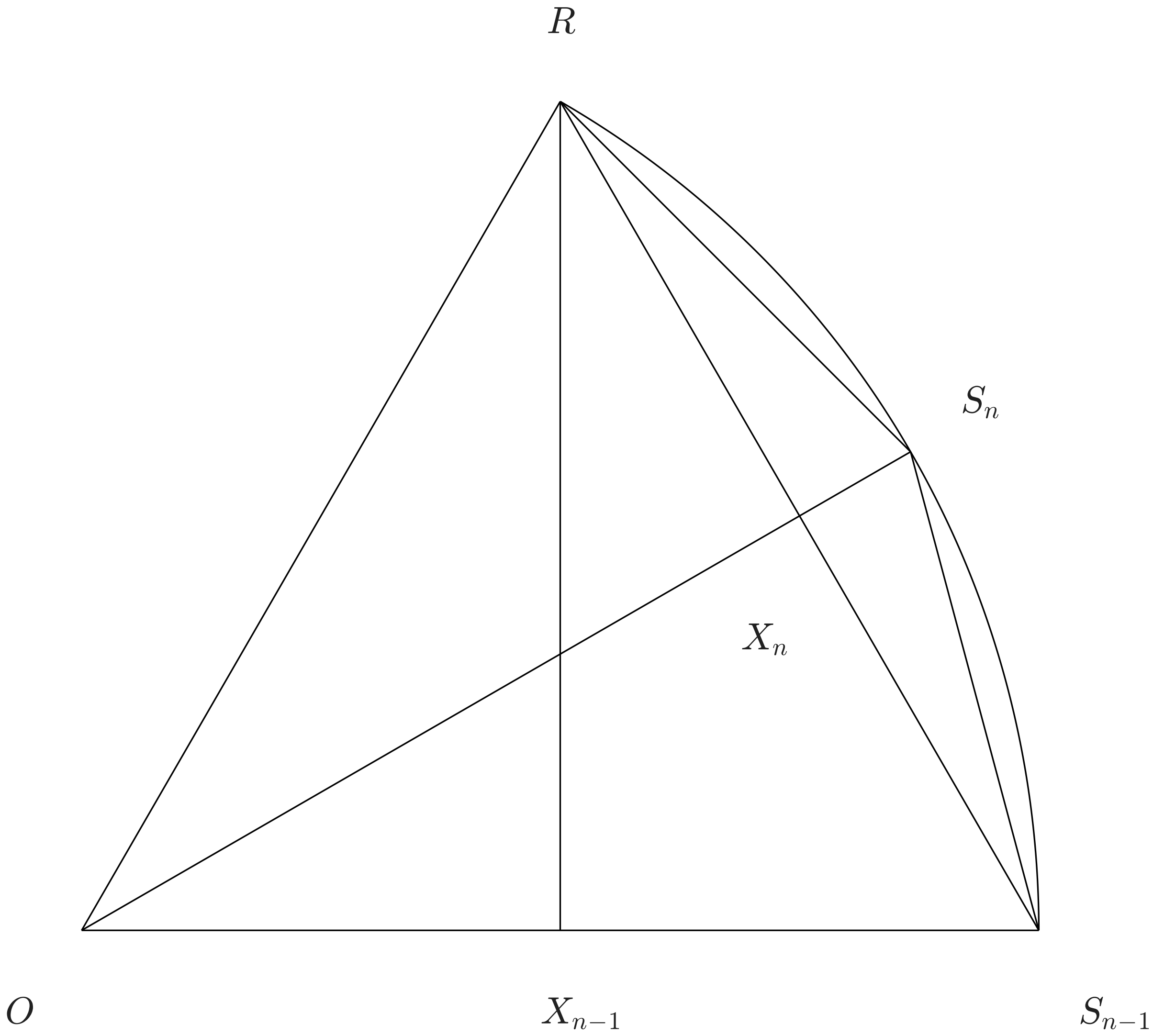
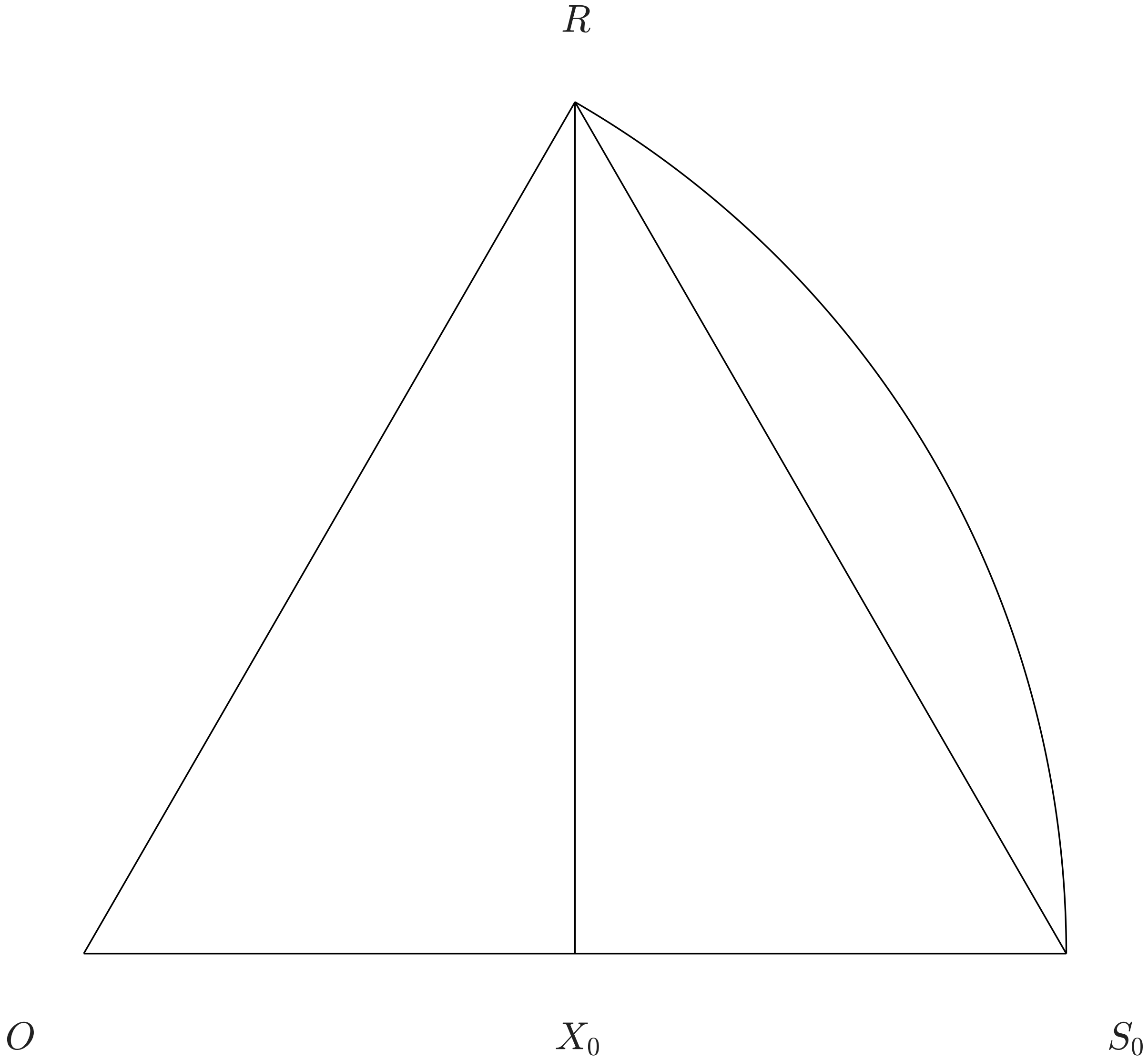


Figure 1: Circular sector.



(a) Recursive step.

Figure 2: Circular sector sequence.



(b) Base case.

Figure 2: Circular sector sequence.

$$F = ma = -kx \tag{224}$$

$$ma = -kx \tag{225}$$

$$x(t) = \cos(r, \omega t) \tag{226}$$

$$a(t) = \frac{d^2 x}{dt^2} \tag{227}$$

$$a(t) = \frac{d^2 \cos(r, \omega t)}{dt^2} \tag{228}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2} \tag{229}$$

$$a(x) = \frac{-\omega^2 x}{r^2} \tag{230}$$

$$m\frac{-\omega^2 x}{r^2} = -kx \tag{231}$$

$$\frac{m(-\omega^2 x)}{r^2} = -kx \tag{232}$$

$$\frac{-m\omega^2 x}{r^2} = -kx \tag{233}$$

$$\frac{-m\omega^2 x}{r^2} = \frac{-kr^2 x}{r^2} \tag{234}$$

$$m\omega^2 = kr^2 \tag{235}$$

$$v(t) = \frac{dx}{dt} \tag{236}$$

$$v(t) = \frac{d\cos(r, \omega t)}{dt} \tag{237}$$

$$v(t) = \frac{-\omega \sin(r, \omega t)}{r} \tag{238}$$

$$K(t) = \frac{mv^2(t)}{2} \tag{239}$$

$$K(t) = \frac{m\left(\frac{-\omega\sin(r,\omega t)}{r}\right)^2}{2} \tag{240}$$

$$K(t) = \frac{m\frac{(-\omega\sin(r,\omega t))^2}{r^2}}{2} \tag{241}$$

$$K(t) = \frac{m\frac{(-\omega)^2\sin^2(r,\omega t))^2}{r^2}}{2} \tag{242}$$

$$K(t) = \frac{m\frac{\omega^2\sin^2(r,\omega t)}{r^2}}{2} \tag{243}$$

$$K(t) = \frac{\frac{m\omega^2\sin^2(r,\omega t)}{r^2}}{2} \tag{244}$$

$$K(t) = \frac{\frac{kr^2\sin^2(r,\omega t)}{r^2}}{2} \tag{245}$$

$$K(t) = \frac{k\sin^2(r, \omega t)}{2} \tag{246}$$

$$U(t) = \frac{kx^2(t)}{2} \tag{247}$$

$$U(t) = \frac{k\cos^2(r, \omega t)}{2} \tag{248}$$

$$E = K + U \tag{249}$$

$$E = \frac{k\sin^2(r, \omega t)}{2} + \frac{k\cos^2(r, \omega t)}{2} \tag{250}$$

$$E = \frac{k\sin^2(r, \omega t) + k\cos^2(r, \omega t)}{2} \tag{251}$$

$$E = \frac{k(\sin^2(r, \omega t) + \cos^2(r, \omega t))}{2} \tag{252}$$

$$E = \frac{kr^2}{2} \tag{253}$$

$$E = \frac{m\omega^2}{2} \tag{254}$$

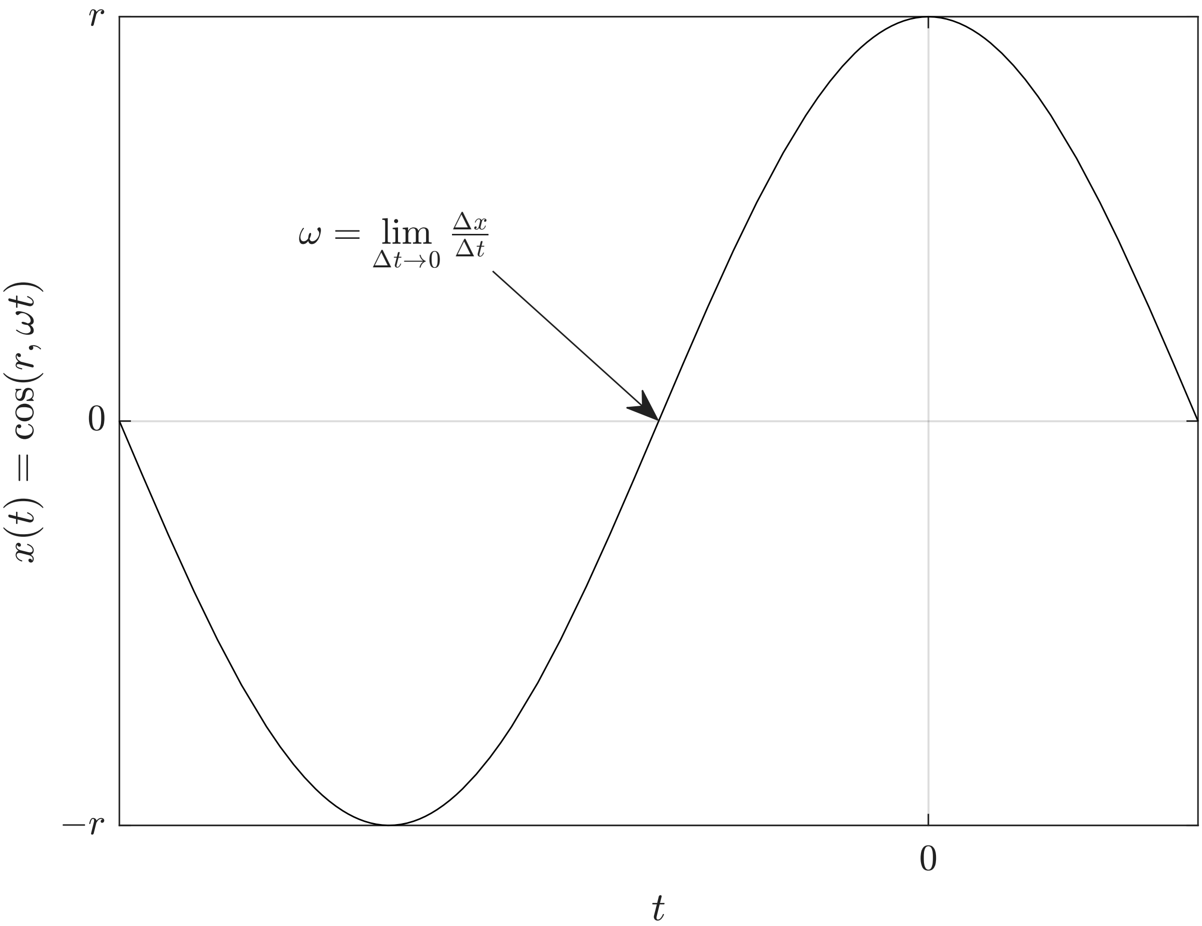


Figure 3: Simple harmonic motion.

$$F = ma \tag{255}$$

$$\omega = \frac{\Delta s}{\Delta t} \tag{256}$$

$$\vec{r_x}(t) = \cos(r, \omega t) \tag{257}$$

$$\vec{r_y}(t) = \sin(r, \omega t) \tag{258}$$

$$\vec{r} = (\vec{r_x}, \vec{r_y}) \tag{259}$$

$$\vec{r}(t) = (\cos(r, \omega t), \sin(r, \omega t)) \tag{260}$$

$$\vec{v} = \frac{d\vec{r}}{dt} \tag{261}$$

$$\vec{v}(t) = \frac{d(\cos(r, \omega t), \sin(r, \omega t))}{dt} \tag{262}$$

$$\vec{v}(t) = \left(\frac{d\cos(r, \omega t)}{dt}, \frac{d\sin(r, \omega t)}{dt}\right) \tag{263}$$

$$\vec{v}(t) = \left(\frac{-\omega\sin(r, \omega t)}{r}, \frac{\omega\cos(r, \omega t)}{r}\right) \tag{264}$$

$$\vec{v}(t) = \frac{\omega\left(-\sin(r, \omega t), \cos(r, \omega t)\right)}{r} \tag{265}$$

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} \tag{266}$$

$$\vec{a}(t) = \frac{d^2(\cos(r, \omega t), \sin(r, \omega t))}{dt^2} \tag{267}$$

$$\vec{a}(t) = \left(\frac{d^2\cos(r, \omega t)}{dt^2}, \frac{d^2\sin(r, \omega t)}{dt^2}\right) \tag{268}$$

$$\vec{a}(t) = \left(\frac{-\omega^2\cos(r, \omega t)}{r^2}, \frac{-\omega^2\sin(r, \omega t)}{r^2}\right) \tag{269}$$

$$\vec{a}(t) = \frac{-\omega^2\left(\cos(r, \omega t), \sin(r, \omega t)\right)}{r^2} \tag{270}$$

$$\vec{a} = \frac{-\omega^2\vec{r}}{r^2} \tag{271}$$

$$\vec{F} = m\vec{a} \tag{272}$$

$$\vec{F} = m\frac{-\omega^2\vec{r}}{r^2} \tag{273}$$

$$\vec{F} = \frac{m(-\omega^2\vec{r})}{r^2} \tag{274}$$

$$\vec{F} = \frac{-m\omega^2\vec{r}}{r^2} \tag{275}$$

$$v = |\vec{v}| \tag{276}$$

$$v = \left|\frac{\omega\left(-\sin(r, \omega t), \cos(r, \omega t)\right)}{r}\right| \tag{277}$$

$$v = \frac{|\omega\,(-\sin(r,\omega t),\cos(r,\omega t))|}{|r|} \tag{278}$$

$$v = \frac{|\omega|\,|(-\sin(r,\omega t),\cos(r,\omega t))|}{|r|} \tag{279}$$

$$v = \frac{\omega\,|(-\sin(r,\omega t),\cos(r,\omega t))|}{|r|} \tag{280}$$

$$v = \frac{\omega\sqrt{(-\sin(r,\omega t))^2+\cos^2(r,\omega t)}}{|r|} \tag{281}$$

$$v = \frac{\omega\sqrt{\sin^2(r,\omega t)+\cos^2(r,\omega t)}}{|r|} \tag{282}$$

$$v = \frac{\omega\sqrt{r^2}}{|r|} \tag{283}$$

$$v = \frac{\omega\,|r|}{|r|} \tag{284}$$

$$v = \omega \tag{285}$$

$$E = \frac{mv^2}{2} \tag{286}$$

$$E = \frac{m\omega^2}{2} \tag{287}$$

$$a = |\vec{a}| \tag{288}$$

$$a = \left|\frac{-\omega^2\vec{r}}{r^2}\right| \tag{289}$$

$$a = \frac{|-\omega^2\vec{r}|}{|r^2|} \tag{290}$$

$$a = \frac{|-\omega^2|\,|\vec{r}|}{|r^2|} \tag{291}$$

$$a = \frac{\omega^2\,|\vec{r}|}{|r^2|} \tag{292}$$

$$a = \frac{\omega^2 r}{|r^2|} \tag{293}$$

$$a = \frac{\omega^2 r}{r^2} \tag{294}$$

$$a = \frac{\omega^2}{r} \tag{295}$$

$$F = \frac{m\omega^2}{r} \tag{296}$$

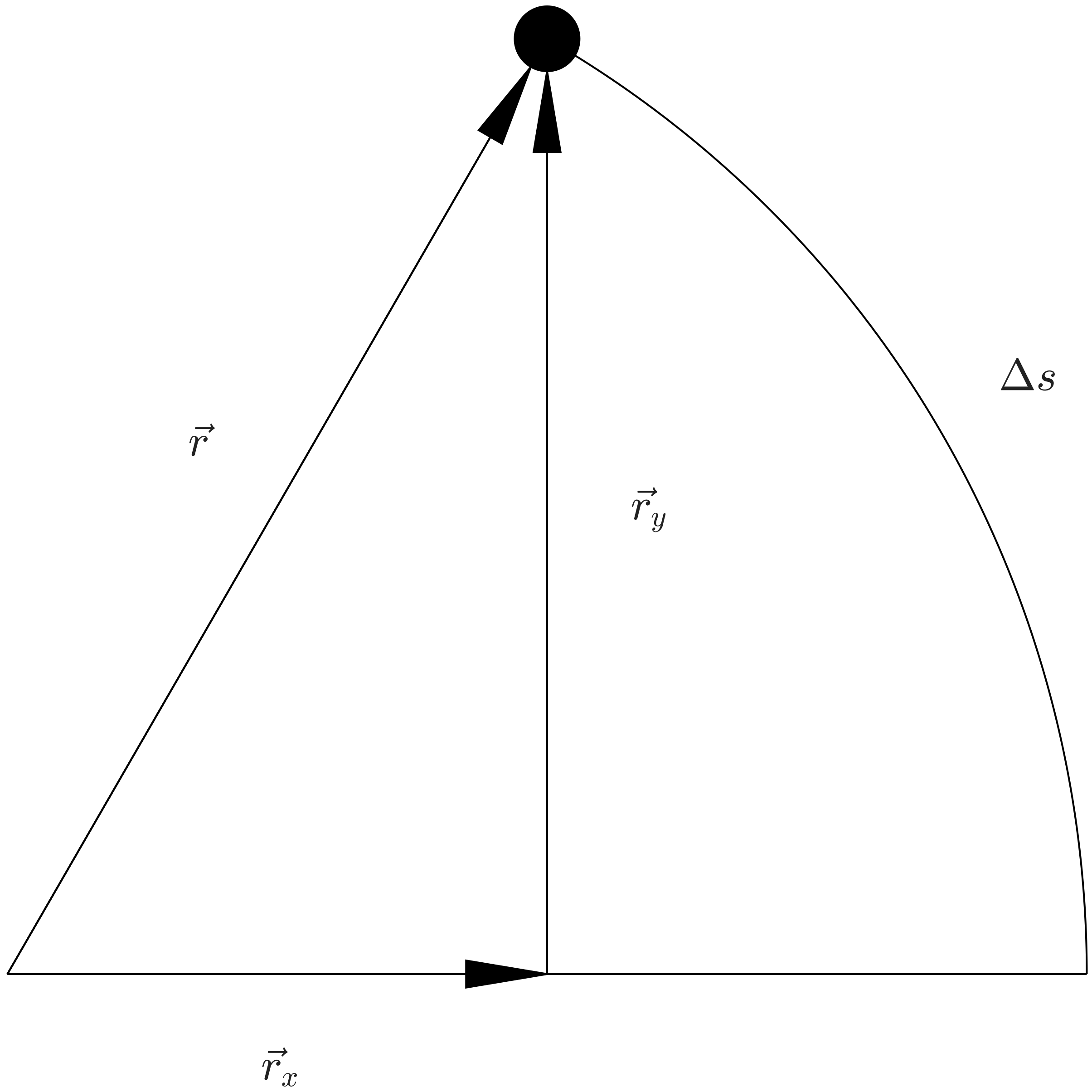


Figure 4: Uniform circular motion.